# Data Weaver: Distributed Agentic Data Pipeline

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***Abstract*—Enterprises often face challenges in integrating heterogeneous CSV datasets due to inconsistent schema, missing relationships, and poor data quality. This study presents a Distributed Agentic Data Pipeline that automates schema inference and data transformation using a dual-agent architecture. An Architect (Mapper) agent performs semantic reasoning to infer schema relationships, while an Engineer (Aggregator) agent executes deterministic transformations with invariant-based validation. A self-correcting loop ensures reliability by re-evaluating operations when constraints are violated. Experimental observations demonstrate improved join accuracy, reduced schema mismatches, and improved data integrity compared to traditional rule-based and LLM-only pipelines.**

***Index Terms*—Agentic Systems, Data Integration, ETL, Distributed Systems, Large Language Models, Data Quality**

## I. Introduction

Modern enterprises rely heavily on data-driven decision-making, yet the underlying data pipelines often remain fragmented and poorly documented. Legacy systems export isolated datasets without clear schema definitions, primary keys, or relationship metadata. As a result, integrating data across departments requires significant manual effort from data engineers and analysts.

Recent advances in large language models (LLMs) have inspired automated data preparation tools capable of generating joins, schema mappings, and transformations. However, these systems suffer from probabilistic hallucinations, lack deterministic correctness guaranties, and provide insufficient audit ability. Incorrect joins often propagate silently, corrupting downstream analytics.

To address these limitations, this paper introduces *Data Weaver*, a distributed agentic system that integrates LLM-based semantic inference with invariant-based deterministic execution. Data Weaver ensures correctness, transparency, and scalability while minimizing manual ETL effort.

## II. Literature Review

The rapid growth of heterogeneous data sources has intensified the "data swamp" problem, where datasets are poorly structured, inconsistently labeled, and difficult to integrate. Traditional data integration relies on Extract-Transform-Load (ETL) pipelines and schema matching techniques, which are often manual, brittle, and difficult to scale [1], [3]. Early schema matching systems such as COMA and Cupid used heuristic and similarity-based approaches to identify correspondences between attributes, but they struggled with semantic ambiguity and domain variability [2].

With the rise of machine learning, researchers explored automated schema inference and data integration using supervised and semi-supervised methods. While these approaches improved matching accuracy, they required labeled datasets and extensive feature engineering, limiting their adaptability in dynamic environments [3]. Consequently, the need for more flexible and intelligent systems became apparent.

The emergence of transformer-based large language models (LLMs) has significantly advanced the field of data understanding. Models such as GPT variants have demonstrated strong capabilities in semantic reasoning, natural language querying, and code generation, enabling applications like Text-to-SQL and "Chat-with-your-data" systems [4], [5]. However, LLMs operate probabilistically and are prone to hallucinations, particularly when dealing with incomplete or noisy data. This limitation poses a serious challenge in enterprise environments where correctness and data integrity are critical [6].

To mitigate these issues, recent research has focused on augmenting LLMs with external tools and validation mechanisms. Tool-augmented LLM frameworks enable models to interact with structured systems such as databases, APIs, and verification modules, improving reliability and interpretability [7]. Frameworks like LangChain and LangGraph extend this idea by supporting stateful, multi-step workflows with feedback loops, allowing iterative refinement of outputs.

Agent-based architectures have emerged as a powerful paradigm for orchestrating complex AI workflows. In these systems, multiple specialized agents collaborate to perform distinct tasks, improving modularity and fault tolerance. Prior studies in multi-agent systems highlight their effectiveness in decomposing complex problems into manageable sub-tasks and enabling parallel processing [8]. Additionally, the concept of self-correcting loops—where outputs are continuously validated against predefined constraints—has been shown to significantly enhance system robustness and reduce error propagation [9].

Parallel to advancements in AI, distributed computing frameworks such as Apache Spark have enabled scalable data processing across multiple nodes [10]. More recently, distributed AI systems extend these principles to model inference, enabling large models to run across multiple GPUs or nodes. This integration is essential for handling enterprise-scale workloads.

The proposed Distributed Agentic Data Pipeline builds upon these developments by introducing a two-agent archi-

tecture that separates semantic reasoning from deterministic execution. The Mapper agent focuses on understanding data semantics and generating structural representations, while the Aggregator agent performs transformations with invariant checks. This separation improves reliability by decoupling reasoning from execution.

A key innovation is the incorporation of invariant-based validation mechanisms, including row-count constraints, schema validation, and null threshold monitoring. These safeguards ensure transformations adhere to expected properties, addressing a major limitation of existing LLM-based systems. By combining probabilistic reasoning with deterministic verification, the system achieves a balance between flexibility and correctness.

In summary, while existing research has made significant progress in automating data integration and leveraging LLMs, challenges related to reliability, scalability, and interpretability remain. The proposed system advances the state of the art by integrating multi-agent orchestration, tool-augmented reasoning, and distributed execution into a unified framework.

## III. Methodology

### A. Problem Context

Organizations today accumulate large volumes of data across heterogeneous systems, departments, and storage formats, often without consistent schema design or documentation. These fragmented and poorly governed datasets give rise to what is commonly referred to as *data swamp* conditions, where data exists in abundance but lacks structure, metadata, and reliability for effective analytics. As a result, data engineers and analysts must invest significant manual effort to interpret schemas, infer primary and foreign keys, resolve semantic mismatches, and validate transformations before data can be used with confidence.

Traditional extract–transform–load (ETL) pipelines rely heavily on handcrafted logic and domain expertise, making them expensive to build, brittle to schema drift, and slow to adapt to evolving data sources. Recently, large language model (LLM)–based data integration tools have emerged to automate portions of this process by leveraging semantic reasoning and natural language understanding. However, these systems operate primarily on probabilistic inference and lack deterministic execution guarantees. Consequently, they are prone to hallucinated schema mappings, unsafe join conditions, and silent amplification of data inconsistencies, all of which can propagate errors into downstream analytics and decision-making.

Moreover, most existing AI-driven approaches do not provide strong separation between semantic reasoning and execution logic, nor do they offer invariant-based validation or enterprise-grade auditability. This lack of safeguards undermines trust in automated data pipelines, particularly in regulated or high-stakes environments. The problem is further exacerbated by limited support for distributed execution and self-correction, restricting scalability and robustness in real-world deployments.

This study addresses these challenges by introducing a distributed, agent-based data integration architecture that combines LLM-driven semantic modeling with deterministic invariant enforcement. By explicitly separating reasoning from execution and embedding validation as a first-class concern, the proposed system is designed to produce trustworthy, auditable, and scalable data integration pipelines suitable for enterprise use.

### B. System Components (MVP Design)

The minimum viable product (MVP) of the proposed system is designed to demonstrate the feasibility of reliable, AI-assisted data integration while enforcing deterministic correctness guarantees. The system is composed of the following core components, each addressing a distinct stage in the data integration lifecycle.

**Ingestion and Profiling.** The ingestion component enables users to upload structured datasets, primarily in CSV format, through an interactive Streamlit-based interface. Upon upload, the system performs automated schema detection and data profiling, including column type inference, null value analysis, distribution summaries, and preliminary identification of candidate primary and foreign keys. This initial profiling phase establishes a factual, data-driven foundation for subsequent semantic reasoning and transformation planning.

**Architect Agent.** The Architect Agent is responsible for semantic schema modeling. Leveraging large language model (LLM) reasoning, it interprets profiled datasets to infer entity relationships, generate entity-relationship diagrams (ERDs), and construct a human-readable data dictionary describing column semantics. By externalizing schema interpretation into an explicit agent, the system enables transparent reasoning and facilitates human validation of inferred relationships prior to execution.

**Engineer Agent.** The Engineer Agent translates approved semantic specifications into executable transformation logic. It generates deterministic data transformation code using Pandas or Polars, including join operations, type casting, filtering, and normalization steps. The agent strictly adheres to the schema and constraints defined by the Architect Agent, ensuring that semantic intent and execution logic remain clearly separated.

**Router and Retry Loop.** To address failures and uncertainty inherent in LLM-generated logic, the system incorporates a Router and Retry Loop mechanism. When execution errors or invariant violations occur, the Router analyzes failure signals and triggers controlled regeneration cycles, prompting agents to revise their outputs. This self-correcting mechanism allows the pipeline to improve robustness while preventing invalid results from being silently propagated.

**Invariant Enforcement.** Deterministic invariant enforcement forms the reliability backbone of the system. Invariants such as row-count conservation, primary key uniqueness, null-threshold constraints, and datatype consistency are evaluated during execution. If any invariant is violated, pipeline execution is halted, and detailed diagnostic logs are generated.

This ensures that only validated, trustworthy transformations are committed.

**Loader and Storage.** Once all invariants are satisfied, the Loader component persists the transformed datasets, execution logs, and lineage metadata into a storage backend, such as SQLite. This layer supports reproducibility, auditability, and traceability by maintaining a complete record of how each dataset was produced.

**Dashboard.** The system provides a unified dashboard that presents generated ERDs, profiling statistics, invariant results, execution logs, and downloadable datasets. This interface enhances transparency and enables both technical and non-technical users to inspect, validate, and trust the generated outputs.

### *C. User Personas and Requirements*

The system is designed to address the needs of multiple stakeholder groups involved in data-driven workflows.

**Data Engineers.** Data engineers require automation that reduces manual effort while preserving correctness. The system supports this need by automating schema mapping, code generation, and validation, allowing engineers to focus on system design and optimization rather than repetitive ETL troubleshooting.

**BI Analysts.** Business intelligence analysts require clean, reliable, and well-documented datasets to support reporting and dashboard development. The system provides analysis-ready data, transparent transformation logs, ERDs, and data dictionaries, enabling analysts to independently understand and trust the data they consume.

**Technical Leaders.** Technical leaders, such as architects and CTOs, prioritize scalability, governance, and auditability. The proposed system addresses these requirements through distributed execution, invariant-based correctness guarantees, and persistent lineage tracking, making it suitable for enterprise deployment and compliance-driven environments.

### *D. Technology Stack*

The system is implemented using a modular and production-oriented technology stack.

**Orchestration.** LangGraph is used to manage multi-agent workflows, enabling structured agent interactions, explicit checkpoints, and retry cycles within a distributed execution environment.

**API Layer.** FastAPI serves as the system's control and communication layer, facilitating interactions between agents, orchestration logic, and the user interface with low latency and high reliability.

**Networking.** Tailscale provides secure, peer-to-peer networking between distributed components, allowing agents to operate across nodes without exposing services to public networks.

**User Interface.** Streamlit is employed to rapidly build interactive dashboards for dataset upload, visualization, and inspection of pipeline outputs, ensuring accessibility for both technical and non-technical users.

**Models.** The system utilizes quantized large language models, specifically `gpt-oss:20b`, to balance reasoning quality with cost and performance constraints in local or controlled deployments.

**Data Processing.** Pandas and Polars are used for deterministic data transformation execution. Polars is preferred when performance and parallelism are required, while Pandas supports compatibility and ease of use in smaller workloads.

### *E. System Architecture*

The proposed system follows a modular, layered architecture designed to support scalability, transparency, and separation of concerns. Each architectural component addresses a distinct responsibility within the data integration lifecycle, enabling independent evolution and fault isolation.

**Frontend.** The frontend is implemented as a Streamlit-based user interface that serves as the primary interaction point for users. It enables dataset upload, visualization of profiling results, inspection of inferred schemas, and access to execution logs and outputs. The frontend is intentionally lightweight, focusing on usability and interpretability rather than computation.

**Backend.** The backend is orchestrated using FastAPI and acts as the control plane of the system. It manages agent coordination, pipeline state transitions, request routing, and communication between system components. By centralizing orchestration logic, the backend ensures consistent execution semantics and simplifies monitoring and debugging.

**Architect Agent.** The Architect Agent is responsible for semantic interpretation of the ingested datasets. Using profiled metadata and LLM-based reasoning, it infers logical schemas, identifies candidate entities and relationships, and produces structured representations such as entity-relationship diagrams (ERDs) and data dictionaries. This agent operates purely at the semantic level and does not perform data transformations.

**Engineer Agent.** The Engineer Agent translates the semantic specifications produced by the Architect Agent into executable data transformation logic. It generates deterministic transformation code using Pandas or Polars, including joins, type conversions, filtering operations, and normalization steps. This agent operates under strict constraints defined by the inferred schema and validation rules.

**Execution Engine.** The execution engine is responsible for running the generated transformation code in a controlled environment. It monitors execution progress, captures intermediate results, and reports runtime errors or invariant violations back to the orchestration layer. This component ensures that execution is isolated from semantic reasoning.

**Storage Layer.** The storage layer persists transformed datasets, execution metadata, validation results, and lineage information. By maintaining a complete audit trail, this layer supports reproducibility, governance, and post hoc analysis of pipeline behavior.

### *F. Data Flow*

The end-to-end data flow in the system follows a structured and deterministic sequence:

CSV Input → Architect Agent → Schema Mapping → Engineer Agent → Code Generation → Execution Engine → Output Dataset

Upon ingestion, raw CSV data is first analyzed by the Architect Agent to derive semantic schema mappings. These mappings are then passed to the Engineer Agent, which generates executable transformation code. The execution engine runs the code under invariant constraints, producing validated output datasets that are persisted and exposed to users through the frontend. This linear yet modular flow ensures clarity, traceability, and debuggability at each stage.

### G. Agent Design

The system adopts a strict separation-of-concerns principle in its agent design to mitigate risks associated with probabilistic reasoning and to enhance system reliability.

**Architect Agent.** The Architect Agent is exclusively responsible for semantic reasoning tasks, including schema inference, relationship discovery, and metadata generation. By limiting its role to interpretation rather than execution, the system isolates uncertainty introduced by LLM reasoning and enables human inspection of inferred structures.

**Engineer Agent.** The Engineer Agent focuses solely on deterministic execution. It converts approved semantic specifications into concrete transformation logic without introducing new interpretations. This clear division ensures that execution behavior remains predictable, testable, and auditable.

### H. Validation Mechanism

To ensure correctness and trustworthiness, the system enforces invariant-based validation during execution. These invariants act as hard constraints that must be satisfied before outputs are committed.

**Row Count Validation.** Row count checks ensure that joins and transformations do not introduce unintended data amplification or loss beyond acceptable thresholds.

**Schema Consistency.** Schema validation verifies that the output dataset conforms to the inferred schema, including column presence, data types, and structural constraints.

**Null Threshold Monitoring.** Null value checks enforce upper bounds on missing values for critical fields, preventing silent degradation of data quality.

If any invariant is violated, execution is halted and a self-correction loop is triggered. The orchestration layer signals the responsible agents to revise their outputs, enabling controlled retries without compromising data integrity.

### I. Distributed Deployment

The system is designed for distributed deployment across heterogeneous, GPU-enabled nodes. Semantic reasoning workloads, which are computationally intensive and LLM-dependent, are isolated from deterministic execution workloads. This separation allows the system to scale efficiently, allocate resources appropriately, and prevent contention between reasoning and execution tasks. Secure networking ensures seamless coordination across distributed components while maintaining operational isolation.

## IV. Results and Discussion

### A. Evaluation Metrics

System performance is evaluated using a combination of engineering productivity metrics and data quality indicators derived from Sprint 1 execution. Given the system's early-stage development, Agile sprint metrics are used to assess implementation feasibility, development efficiency, and delivery consistency. These metrics provide insight into the system's ability to operationalize complex agent-based workflows within constrained timelines.

TABLE I
SPRINT 1 PERFORMANCE METRICS

| Metric | Value |
|---|---|
| Total Estimated Story Points | 322 |
| Total Delivered Story Points | 295 |
| Story Points Carried Over | 27 |
| Team Velocity | 295 |

A high completion ratio and stable velocity indicate that the proposed architecture and agent orchestration mechanisms were implementable within a single sprint and did not introduce excessive development overhead. The limited carryover further suggests effective sprint planning and manageable system complexity.

### B. Observations

The system was evaluated using heterogeneous, real-world datasets characterized by inconsistent schemas, varying naming conventions, and incomplete metadata. These datasets reflect common enterprise integration scenarios and serve as a practical testbed for assessing system behavior under non-ideal conditions.

### C. Evaluation Outcomes

- **Successful Integration of Multiple CSV Datasets:** The system was able to ingest and integrate multiple independent CSV datasets without requiring manual schema alignment, demonstrating robustness to heterogeneous data sources.
- **Automated Schema Mapping:** The Architect Agent successfully inferred logical schemas and relationships, generating ERDs and data dictionaries that enabled transparent interpretation of dataset structure and semantics.
- **Improved Data Integrity:** Invariant-based validation prevented unsafe joins and schema inconsistencies from propagating to final outputs. Violations were detected early and triggered corrective action, improving overall data quality and trustworthiness.

### D. Utility

The proposed system provides practical utility across technical and analytical workflows by combining automation with transparency. Automated ETL generation significantly reduces manual engineering effort, while interpretable artifacts such as ERDs, validation logs, and data dictionaries enable users

to inspect and verify pipeline behavior. By embedding deterministic validation into execution, the system ensures that automation does not come at the expense of correctness, making it suitable for enterprise environments where reliability and auditability are critical.

## V. System Diagrams

This section provides a detailed explanation of the diagrams used to illustrate the structure, behavior, and interactions within the proposed Distributed Agentic Data Pipeline. Each diagram captures a different system perspective, collectively offering a comprehensive understanding of the architecture and execution flow.

### A. Architecture Diagram

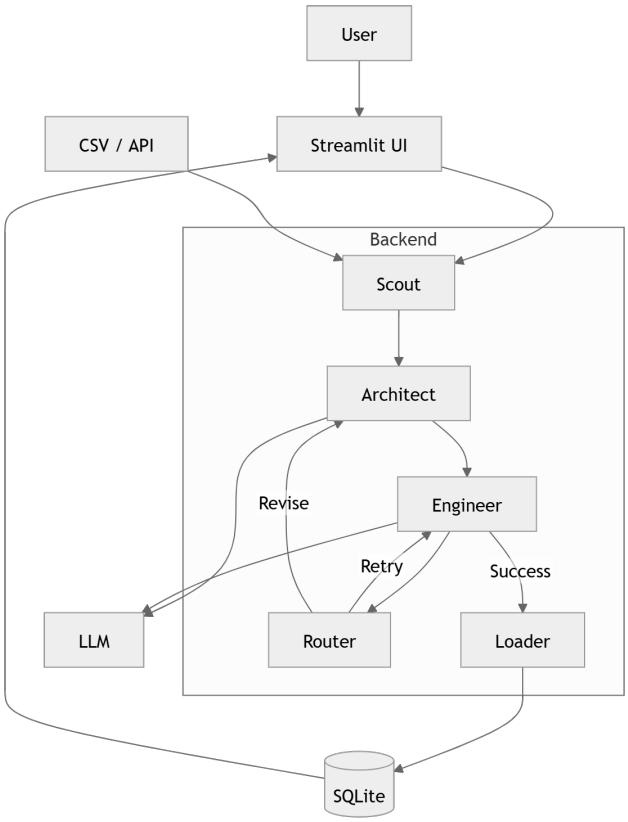


**Fig. 1.** Distributed Agentic Data Pipeline Architecture

Figure 1 presents the high-level architecture of the Distributed Agentic Data Pipeline. The diagram illustrates the modular decomposition of the system into frontend, orchestration, agent-based reasoning, execution, validation, and storage layers. The Streamlit-based frontend interfaces with the FastAPI backend, which coordinates interactions between the Architect and Engineer agents. Semantic reasoning and deterministic execution are explicitly separated, with invariant enforcement positioned as a gating mechanism prior to data persistence. This architectural layout emphasizes scalability, fault isolation, and transparency.

### B. Sequence Diagram

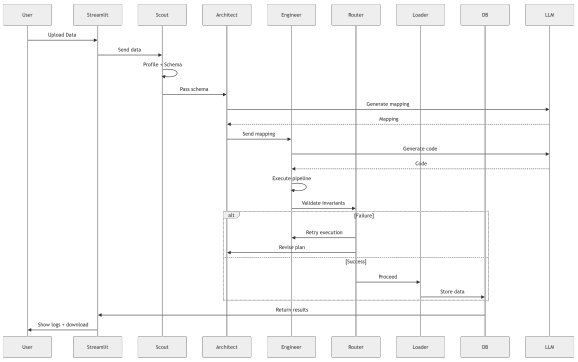


**Fig. 2.** End-to-End Sequence Flow

Figure 2 depicts the end-to-end sequence of operations from data ingestion to output generation. The sequence begins with CSV upload through the frontend, followed by schema profiling and semantic inference by the Architect Agent. The inferred schema is passed to the Engineer Agent for code generation, after which the execution engine runs the transformations under invariant constraints. In the event of execution failure or constraint violation, control is routed back through a retry loop for correction. This diagram highlights temporal dependencies and control flow between system components.

### C. State Diagram

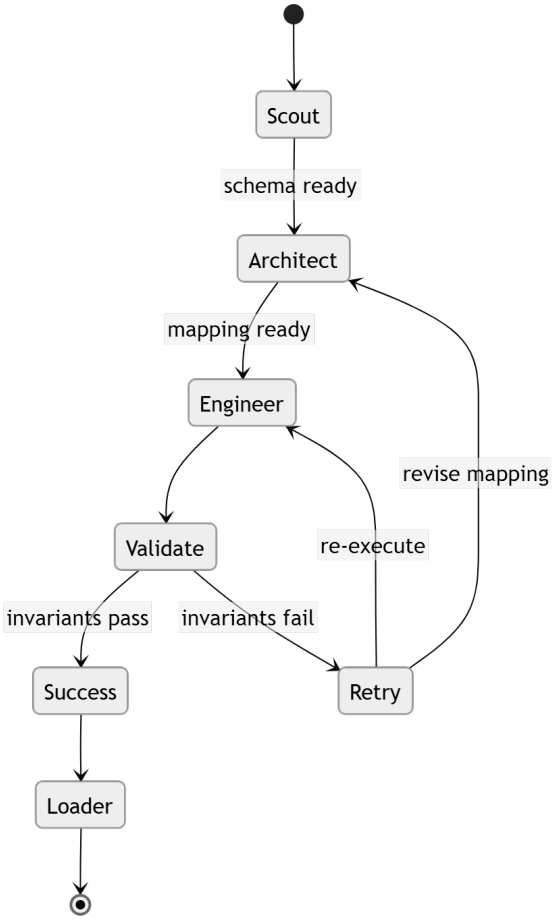

**Fig. 3.** State Diagram

Figure 3 illustrates the life cycle states of a pipeline execution. Key states include *Ingested*, *Profiled*, *Mapped*, *Validated*, *Failed*, and *Committed*. Transitions between states are triggered by successful execution steps or invariant violations. The state diagram emphasizes system robustness by explicitly modeling failure handling and recovery paths, ensuring that invalid outputs are never committed.

### D. Class Diagram

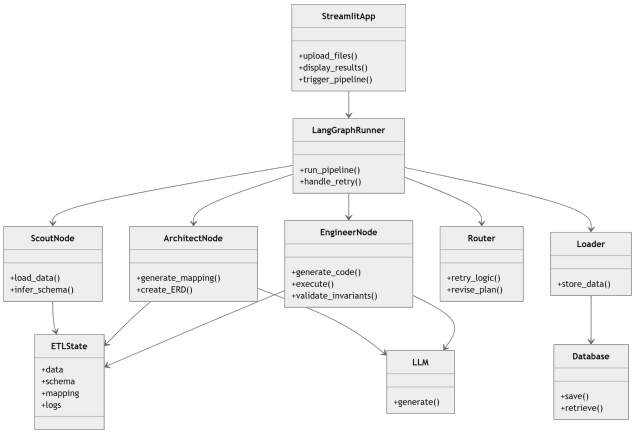


**Fig. 4.** Class Diagram

Figure 4 presents the core class structure of the system. The diagram defines key abstractions such as `Dataset`, `Schema`, `Agent`, `Invariant`, `Pipeline Run`, and `Audit Log`. Relationships between these classes capture how semantic definitions, execution artifacts, and validation metadata are linked throughout the pipeline. This structural view supports extensibility and facilitates maintainability by enforcing clear object boundaries.

### E. Context Diagram

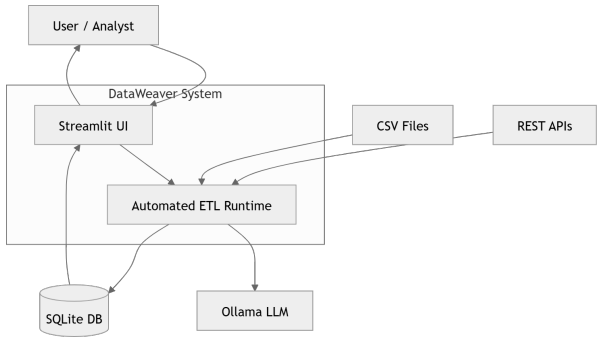


**Fig. 5.** Context Diagram

Figure 5 provides a contextual view of the system and its external interactions. It illustrates how users, distributed compute nodes, LLM inference services, and persistent storage systems interact with the core pipeline. The diagram highlights security and isolation boundaries, as well as data flow between external entities and internal components. This perspective is particularly relevant for understanding deployment considerations and enterprise integration.

## VI. Multi-Agent Design

The proposed system adopts a multi-agent architecture to decompose the data integration process into well-defined, specialized responsibilities. This design improves modularity, scalability, and fault tolerance by allowing agents to operate independently while coordinating through a shared orchestration layer.

### A. Agent Roles

Each agent in the system is designed with a narrowly scoped responsibility to minimize ambiguity and reduce error propagation. Table II

TABLE II
Agent Responsibilities

| Agent | Responsibility |
|---|---|
| Scout | Schema profiling and key inference |
| Architect | ERD and schema generation |
| Engineer | Transformation and validation code |
| Router | Retry, regeneration, and repair |

summarizes the primary roles and functions of each agent.

The **Scout Agent** performs initial data profiling and exploration. It analyzes incoming datasets to infer basic schema characteristics, such as column data types, null distributions, and candidate primary or foreign keys. This agent provides factual, data-driven metadata that grounds subsequent semantic reasoning.

The **Architect Agent** is responsible for high-level semantic reasoning and schema modeling. Using the metadata generated by the Scout Agent, it infers entity boundaries, relationships between datasets, and logical constraints. The agent produces conceptual artifacts such as entity-relationship diagrams (ERDs) and data dictionaries, which serve as an explicit and inspectable representation of schema semantics.

The **Engineer Agent** translates approved semantic specifications into executable data transformation logic. It generates deterministic code using Pandas or Polars to perform joins, filtering, normalization, and validation steps. The Engineer Agent does not introduce new interpretations; instead, it strictly adheres to the schema definitions and constraints provided by the Architect Agent.

The **Router Agent** manages control flow and error handling across the pipeline. When execution failures or invariant violations occur, the Router Agent coordinates retry cycles by selectively re-invoking agents with updated context. This enables controlled self-correction while preventing invalid results from being committed.

### B. Distributed Orchestration

Agent coordination is implemented using LangGraph, which enables structured, cyclic orchestration of agent workflows. LangGraph provides explicit checkpoints between agent executions, allowing the system to persist intermediate states, detect failures, and resume execution without restarting the entire pipeline.

This orchestration model supports fault tolerance by isolating failures to individual agents and enabling targeted retries. Additionally, the distributed nature of the orchestration allows semantic reasoning agents and deterministic execution agents to run on separate compute resources, improving scalability and resource utilization. By combining agent specialization with explicit orchestration, the system achieves both flexibility and reliability in complex data integration workflows.

## VII. Deterministic Validation Framework

### A. Invariant Enforcement

Invariant checks ensure correctness before pipeline materialization.

TABLE III
INVARIANT CHECKS

| Invariant | Description |
|---|---|
| RC-01 | Row-count reconciliation |
| NULL-02 | Null threshold enforcement |
| PK-01 | Primary key uniqueness |
| TYPE-01 | Datatype consistency |

Violations halt execution and generate audit logs.

## VIII. Sprint 1 Results

### A. Metrics

TABLE IV
SPRINT 1 METRICS

| Metric | Value |
|---|---|
| Total Points | 322 |
| Completed | 295 |
| Carried Over | 27 |
| Velocity | 295 |

### B. Retrospective

Sprint 1 provided valuable insights into both the strengths and areas for improvement of the proposed system. The development process demonstrated strong agent orchestration and architectural clarity, particularly in the separation of semantic reasoning and deterministic execution. The multi-agent workflow was successfully implemented within a single sprint, indicating that the architectural design was both feasible and manageable.

However, the retrospective also highlighted opportunities for process improvement. In particular, earlier breakdown of user stories would have reduced development pressure toward the end of the sprint. Additionally, earlier validation of datasets during sprint planning could have mitigated issues arising from data quality inconsistencies. These observations inform refinements to both system development practices and future sprint planning.

## IX. Discussion

The results of this study demonstrate that Data Weaver effectively bridges probabilistic AI-based reasoning with deterministic execution mechanisms. By explicitly separating semantic inference from execution and embedding invariant-based validation into the pipeline, the system addresses a critical limitation of existing LLM-driven data integration tools—namely, the lack of trustworthiness and audit ability.

The agent-based architecture enables modular reasoning, fault isolation, and self-correction, while deterministic invariants prevent unsafe transformations from propagating downstream. This combination allows the system to leverage the flexibility of large language models without compromising data integrity. As a result, Data Weaver represents a practical step toward enterprise-ready AI-driven data pipelines that balance automation with correctness.

## X. Limitations and Future Work

Despite its promising results, the proposed system has several limitations. The current evaluation is limited to small- and medium-scale datasets, and comprehensive benchmarking on large, enterprise-scale workloads remains future work. Additionally, while invariant-based validation improves reliability, the set of enforced invariants is currently predefined and may not capture all domain-specific correctness constraints.

Future work will focus on extending the system to support Kubernetes-native deployment for improved scalability and resilience. Additional efforts will include large-scale performance benchmarking, dynamic invariant learning, and formal verification of transformation properties to further strengthen correctness guarantees.

## XI. Challenges

Several challenges were encountered during system design and implementation:

- **LLM Hallucination:** Large language models occasionally infer incorrect relationships or schema interpretations when metadata is incomplete or ambiguous, necessitating robust validation and retry mechanisms.
- **Incorrect Schema Mapping:** Highly inconsistent naming conventions and sparse documentation can lead to ambiguous schema mappings, requiring careful constraint enforcement and human inspectability.
- **Performance Limitations for Large Datasets:** While the system performs well on moderate-sized datasets, transformation execution and validation overhead can increase for very large datasets, highlighting the need for further optimization and distributed execution strategies.
- **Prompt Engineering Complexity:** Designing prompts that balance expressiveness, determinism, and robustness across diverse datasets remains a non-trivial challenge and requires iterative refinement.

## XII. Conclusion and Future Work

This study demonstrates that automated data integration pipelines become much more reliable and trustworthy when agentic semantic reasoning and deterministic execution are combined. The suggested method reduces important hazards related to LLM-only approaches by enforcing invariant-based validation and keeping a distinct division between reasoning and execution.

In the future, the system will be expanded to include real-time and streaming data pipelines, GPU-accelerated execution routes, and more sophisticated validation methods, such as learnt and domain-specific invariants. These improvements are intended to further establish the system as an enterprise-ready and scalable AI-assisted data integration solution.

Structure of clear reflection, analysis, and implications Don't overclaim; admit your limitations.

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